

CPU

8

Addressing Mode

Addressing Modes

- **The operation field of an instruction specifies the operation to be performed.**
- **This operation must be executed on some data stored in registers or memory word.**

- **The way the operands are chosen during the program execution depends on the addressing mode of the instruction.**
- **Addressing mode is a rule for modifying the address field of the instruction before the operand is actually referenced.**

- Every instructions involves one or more operands. The operand may be processor register, constants or memory word.
- Ex. ADD R1,R2
- AND M[100],R3
- ADD 50 // $AC \leftarrow AC + 50$
- ADD B // $AC \leftarrow AC + B$
- The way an instruction references its operands is called its addressing mode.

➤ There are two modes that need no address fields at all.

➤ These are called the implied and immediate modes.

1. Implied mode

- In this mode, the operands are specified implicitly in the definition of the instruction.
- All register reference instructions that uses accumulator are implied-mode instructions.

➤ Example

Zero-address instruction in a stack-organized computer are implied-mode instructions since the operands are implied to be on the top of the stack.

PUSH A

PUSH B

ADD

2.Immediate Mode

- In this mode the operand is specified in the instruction itself.
- An immediate mode instruction has an operand field rather than an address field.

➤ The operand field contains the actual operand value to be used in conjunction with the operation specified in the instruction.

➤ For e.g. the instruction

ADD 10 // $AC \leftarrow AC + 10$

3.Register Mode

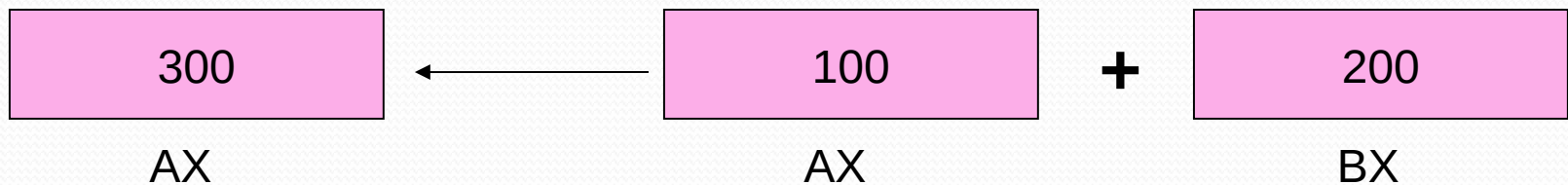
- **When the address field specifies a processor register, the instruction is said to be in the register mode.**
- **In this mode, the operands are in the registers that resides within the CPU.**

- **In general, the register addressing mode is the most efficient because registers are the part of the CPU and no memory reference is need.**

➤ Examples of Register Address mode:

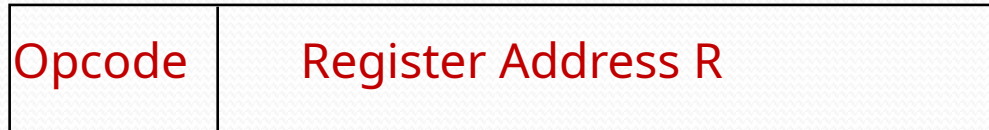
ADD R1 R2 // $R1 \leftarrow R1 + R2$

MUL AX, CX // $AX \leftarrow AX * CX$

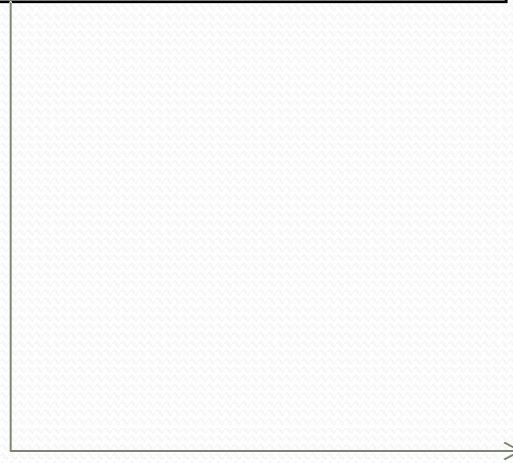
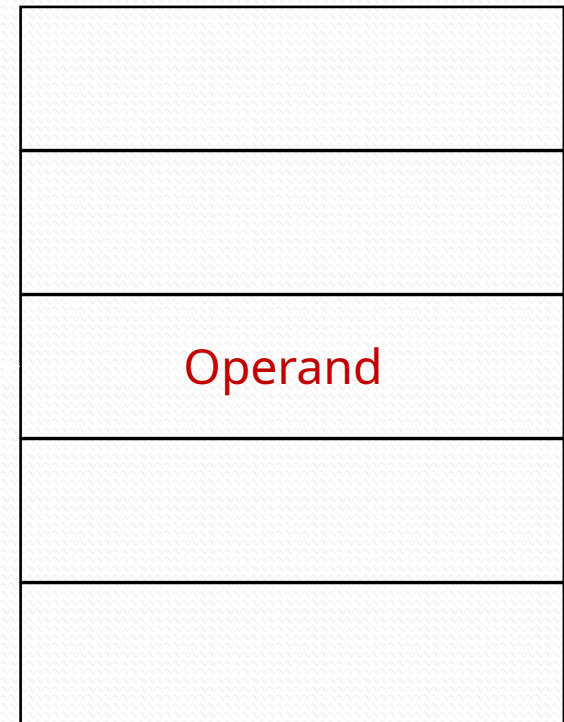


Register Addressing Diagram

Instruction



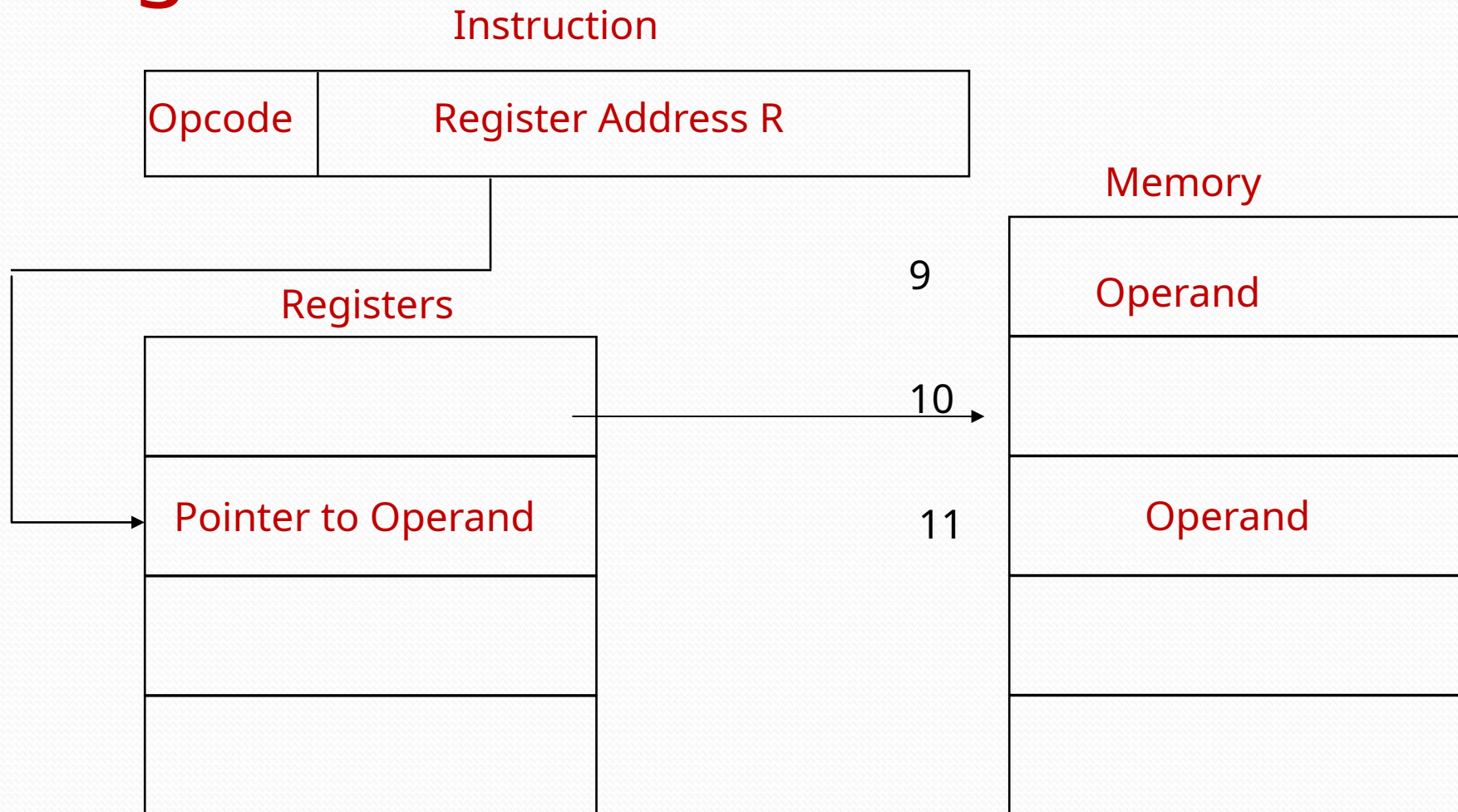
Registers



4. Register Indirect Mode

- In this mode, the instructions specifies a register in the CPU whose contents give the address of the operand in memory.
- In other words, the selected register contains the address of the operand in memory rather than the operand itself.

Register Indirect Addressing Diagram



5. Autoincrement or Autodecrement Mode

- This is similar to register indirect mode except that the register is incremented or decremented after or before its value is used to access memory.



➤ **This is achieved by using the increment and decrement instructions.**

6. Direct Address Mode

- In this mode, the effective address is equal to the address part of the instruction.
- The operand resides in memory and its address is given directly by the address field of the instruction.

7. Indirect Address Mode

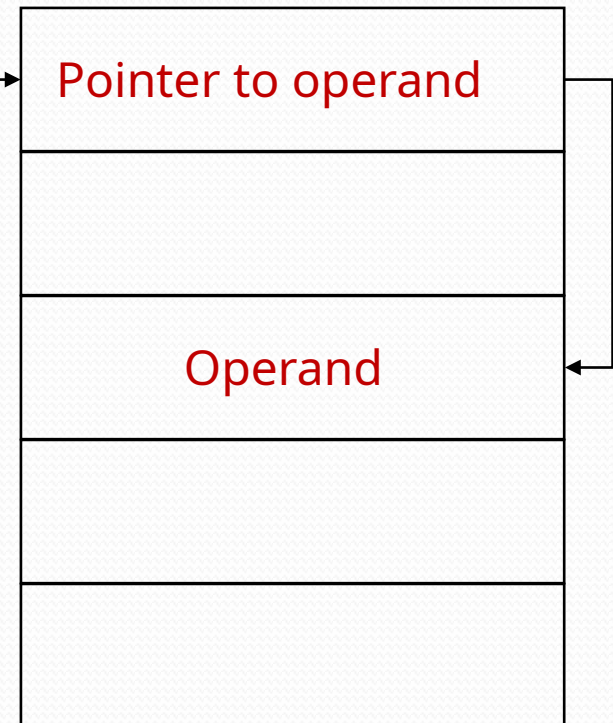
- In this mode the address field of the instruction gives the address where the effective address is stored in memory.
- Control fetches the instruction from the memory and uses its address part to access memory again to read the effective address.

Indirect Addressing Diagram

Instruction



Memory



8. Relative Address Mode

- In this mode the content of the program counter is added to the address part of the instruction in order to get the effective address.

Effective Address = Content of PC +
Address of instruction.

- **The address part of the instruction is usually a signed number (in 2's complement notation) which can either be positive or negative.**
- **When this number is added to the content of PC, the result produces an effective address.**

9.Indexed Addressing Mode

- In this mode the content of an index register is added to the address part of the instruction to obtain the effective address.
- The index register is a special CPU register that contains the index value.

- The address field of the instruction defines the beginning address of a data array in memory.
- The distance between the beginning address and the address of the operand is called the index value stored in the index register.

10.Base Register Addressing Mode

- In this mode the content of a base register is added to the address part of the instruction to obtain the effective address.
- This is similar to the indexed addressing mode except that the register is now called a base register instead of an index register.



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